

Developing India's digital backbone: How digital infrastructure will power India's \$5 trillion economy



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When we think of national infrastructure, our minds instinctively travel to highways winding across mountains, ports buzzing with the movement of cargo, rail networks meandering through vast plains and valleys, and power plants humming along the hinterlands.

However, quietly in the background, a transformative digital infrastructure corridor is rising, comprising fiber optics, datacenters, cloud platforms, 5G wireless, and satellite constellations. India is building its 'fifth pillar' of infrastructure - one that is measured in terabits and latency, compute cycles, and uptime. This pillar is powering a nation as it races to establish itself as a leader in the global market.

The fifth pillar: Beyond roads, rails, ports, and power

The growth story of India over the last few decades has been based on the development of the core infrastructure such as roads and railways to facilitate the movement of goods and people, ports to facilitate trade, and power to facilitate industry. The next wave of productivity is not about the movement of physical products, but the movement

of data, which enables digital services, and building a connected digital ecosystem.

The Ministry of Electronics and IT states that the digital economy's contribution to the country's GDP is 11.74% and the productivity of this sector is much higher compared to that of other sectors in the country. Therefore, digital infrastructure is no longer peripheral but foundational to the country's economic growth.

Cross-industry play: The digital thread interwoven across sectors

Take the example of the financial sector, where networks, cloud, and processing infrastructure have enabled the scaling of digital payments using the Unified Payments Interface (UPI) platform. Similarly, the healthcare industry has been redefined through telemedicine consultations enabled by strong connectivity and the capacity of the datacenters to facilitate these interactions.

For the manufacturing sector, a robust system of sensors, IoT, 5G connectivity, cloud analytics, and high-speed connection to datacenters is essential to enable them to scale their operations by using technology. While roads, ports, rail and power transport physical material and energy, digital infrastructure creates new value in industries and enables innovation.

Digital infrastructure - an economic flywheel

Digital infrastructure in India is fast becoming an economic flywheel. As investments in fiber and datacenters increase, latency reduces and connectivity increases, enabling more companies to switch to cloud, edge computing, 5G, and satellites. This improves productivity and gives rise to new business models and digital services, which could enhance exports, drive economic growth across sectors, and expand the digital ecosystem further, leading to more demand for digital infrastructure in the country.

And investors are taking notice which is evident from the growth of India's datacenter industry, which is estimated to exceed 4.5 GW by 2030, backed by \$25-30 billion in new investments. Meanwhile, 5G and AI are driving the demand for fiber optics, which would need close to twice the fiber density of traditional datacenters in India.

For India's near-term target of becoming a \$5 trillion economy, this shift is crucial as digital infrastructure is no longer a 'nice-to-have' but a priority investment - one that could enhance productivity across sectors.

India's unique opportunity: Strategic imperatives for the industry and policymakers

The Indian economy's growth is supported by several advantages such as the young population, high percentage of smartphone adoption, a favourable geo-economic position, and a strong policy background. These can be important enablers for the country to develop a robust digital infrastructure system. However, India generates nearly 20% of the world data, yet the country is still under-represented in the datacenter capacity.

Industry and policymakers should focus on:

1. Think 'network infrastructure' not just vertical upgrades: Manufacturing plants, hospitals, banks should consider implementing technologies such as digital twin, remote monitoring, and analytics

2. Invest ahead of utilization: High-latency connectivity is a bottleneck. Therefore, it is important to focus on initial deployment of fiber/backhaul, datacenter co-location, and cloud architecture to gain a competitive edge.

3. Public-private collaboration: The government must create the right environment with right-of-way, local permissions, subsidies, or viability gap funding for rural fiber and datacenter zones.

Looking ahead: How digital infrastructure could define the next decade

As India marches toward the \$5 trillion economy milestone, the victors will not be the company with the most factories, the most docks, but that which is able to tap into the digital backbone connecting them. It's not just machines and labour that bring the future factory, but the sensors, networks, cloud analytics, and data flow across the globe contribute to it. Not just doctors, but also by telepresence, patient-data platforms, and real-time analytics, which will enable future hospitals.

Digital infrastructure, therefore, is not just an enabler but a strategic multiplier of growth, inclusion, and innovation.